Week 4

More JCF and Simple Data Structures

School of Computing Technologies
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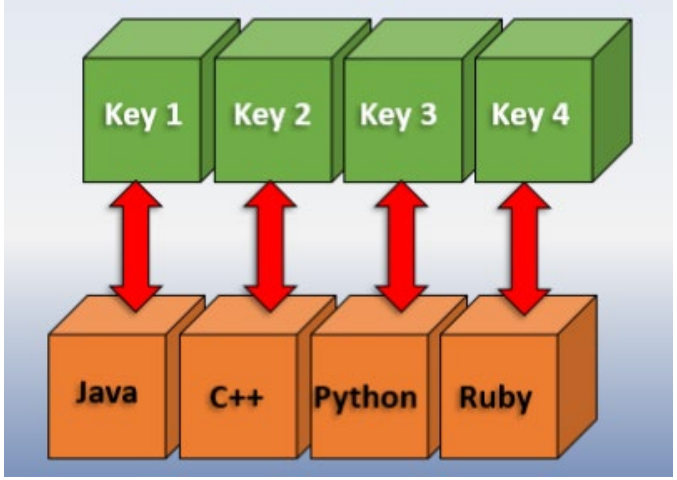
Map

Map stores key-value pairs.

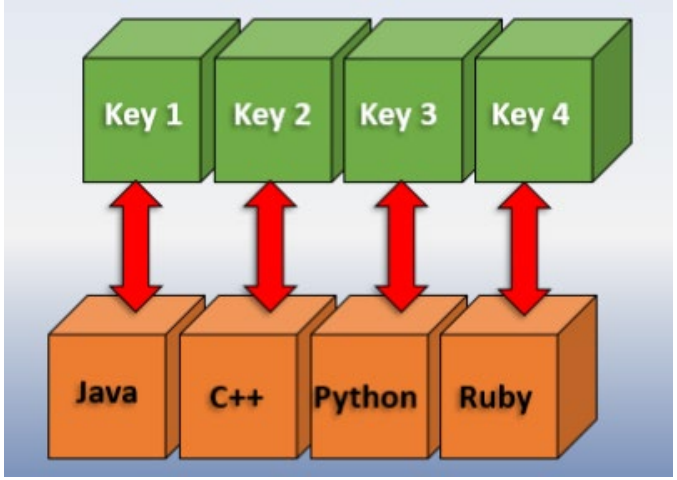
- Keys must be **unique**
- Maps are used for fast retrieval of data



Map

Map stores key-value pairs.

- **HashMap:** efficient for locating a value, inserting/deleting mapping
- **LinkedHashMap:** Extends HashMap with a linked list – elements can be accessed in the order of insertion
- **TreeMap:** Elements are sorted



Map Demo

```java
public class TestMap{
    public static void main(String[] args){
        HashMap hashMap = new HashMap();
        hashMap.put("Charles Theva", new Account("S123",130.0));
        hashMap.put("Bill Cooper", new Account("S124",90.0));
        hashMap.put("Abraham Lincoln", new Account("S126",220.0));
        hashMap.put("Bill Cooper", new Account("S124",120.0));
        System.out.println("Entries in Hashmap");
        displayMap(hashMap);
        System.out.println("\nEntries in Treemap");
        TreeMap treeMap = new TreeMap(hashMap);
        displayMap(treeMap);
    }
    public static void displayMap(Map m){
        Set keySet = m.keySet();
        Iterator iterator = keySet.iterator();
        while (iterator.hasNext()) {
            String key = (String) iterator.next();
            System.out.println(key + ":" + m.get(key) );
        }
    }
}
```

```java
class Account{
    private String ID;
    private double balance;
    public Account(String ID, double balance){
        this.ID = ID;
        this.balance = balance;
    }
    public String getID() { return ID; }
    public double balance() { return balance; }
    public String toString(){
        return new String("ID = "+ ID + " bal = " + balance);
    }
}
```

```
Entries in Hashmap
Bill Cooper:ID = S124 bal = 120.0
Abraham Lincoln:ID = S126 bal = 220.0
Charles Theva:ID = S123 bal = 130.0

Entries in Treemap
Abraham Lincoln:ID = S126 bal = 220.0
Bill Cooper:ID = S124 bal = 120.0
Charles Theva:ID = S123 bal = 130.0
```

Review Question 1

- What is the output of the following program?

```
import java.util.*;
public class MapTest {
   public static void main(String[] args) {
      Map<String, String> accountsMap = new TreeMap<String, String>();
      accountsMap.put("E123", "Charles");
      accountsMap.put("E174", "Matt");
      accountsMap.put("E542", "Brendan");
      accountsMap.put("E174", "Tobin");

      System.out.println(accountsMap);

      System.out.println(accountsMap.keySet());

      for (String en : accountsMap.keySet())
         System.out.println(accountsMap.get(en));
   }
}
```

Review Question 2

Challenge

- Construct the TreeMap to sort entries based on truncated keys, specifically the digits following the initial digit of the original key.

Simple Data Structures

Data structures are designed for managing **a collection of data** more efficiently.

- **Access** – Retrieving a particular element given its position in the collection
- **Search** – Finding a particular element in the collection
- **Insertion** – Adding a new element to the collection
- **Deletion** – Deleting an existing element to the collection

Which data structure is more efficient in access?

Exercise

- Write a program to create an ArrayList of Integer type.
- Add numbers from 0 to 1e5 to the ArrayList
- Print the element located in the middle of the list
- Record the time of finding the element and printing it out.

Which data structure is more efficient in access?

Exercise

- Change the size of the ArrayList from 1e5 to 1e6, and record the time of search
- How about further increasing it to 1e7?

Which data structure is more efficient in access?

Exercise

- Change the size of the ArrayList from 1e5 to 1e6, and record the time of search
- How about further increasing it to 1e7?
- How about changing the data structure to LinkedList?

Which data structure is more efficient in search?

Summary of the access cost*

	1e5	1e6	1e7
ArrayList			
LinkedList			

Which data structure is more efficient in search?

Exercise

- Write a program to create an ArrayList of Integer type.
- Add numbers from 0 to 1e5 to the ArrayList
- Search for “-1” from the list
- Record the time of the search

Which data structure is more efficient in search?

Exercise

- Change the size of the ArrayList from 1e5 to 1e6, and record the time of search
- How about further increasing it to 1e7?

Which data structure is more efficient in search?

Exercise

- Change the data structure from ArrayList to LinkedList
- How about changing it to HashSet?
- Try changing the data structure to HashMap?

Which data structure is more efficient in search?

Summary of the search cost*

	1e5	1e6	1e7
ArrayList			
LinkedList			
HashSet			
HashMap			

Which data structure is more efficient in insertion/deletion?

Exercise

- Write a program to create an ArrayList of Integer type.
- Add numbers from 0 to 1e3 to the ArrayList
- Deleting the first element recursively from ArrayList until there is no element in the list
- Record the time of the insertion + deletion

Which data structure is more efficient in insertion?

Exercise

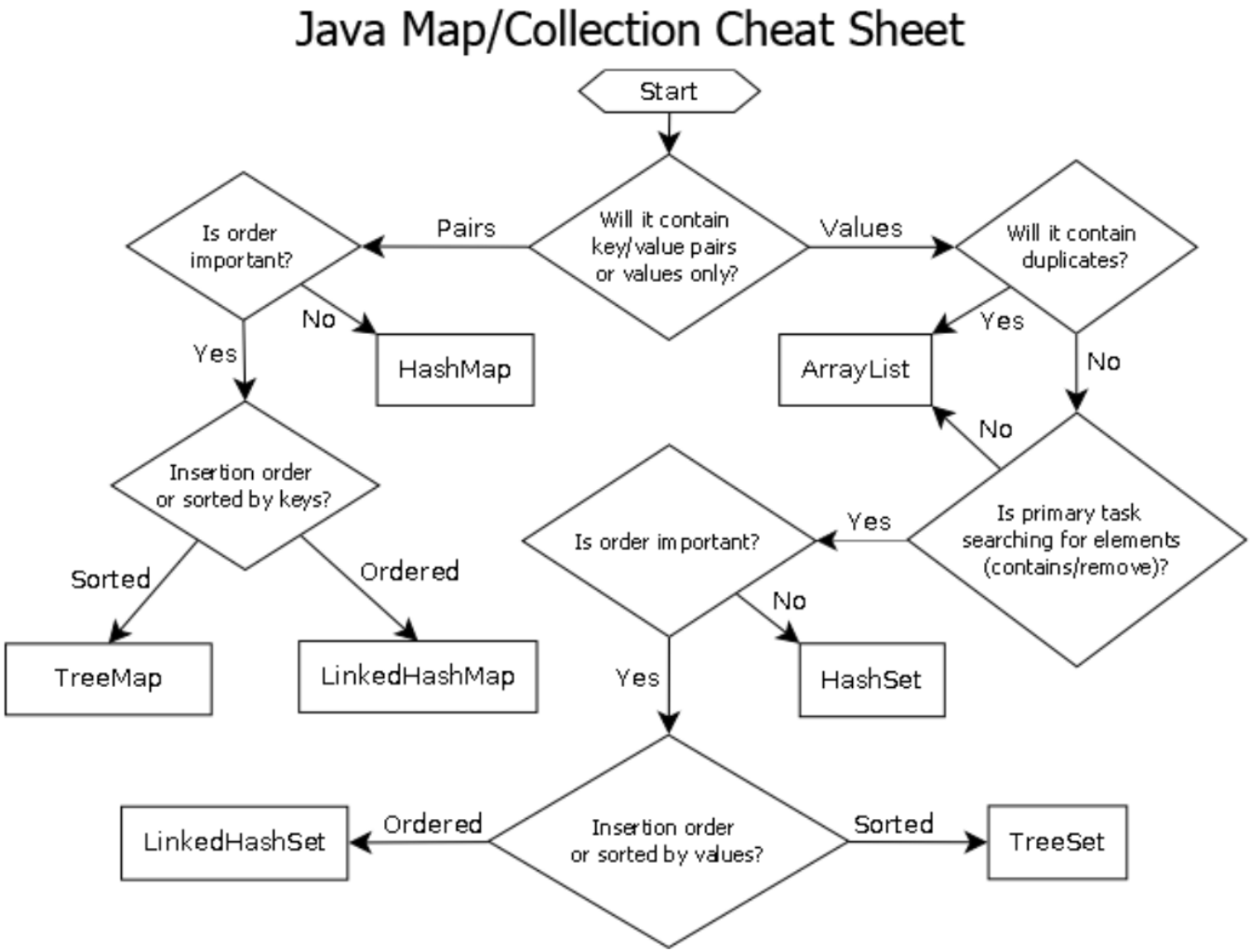
- Vary the size of Array in 1e3, 1e4, and 1e5
- Change the data structure to LinkedList, Set, HashMap

Which data structure is more efficient in insertion?

Summary of the insertion cost*

	1e5	1e6	1e7
ArrayList			
LinkedList			
HashSet			
HashMap			

A Cheat Sheet



http://www.sergiy.ca/guide-to-selecting-appropriate-map-collection-in-java/

Review Question 3

- What is the advantage of using Collections classes (as opposed to building your own)?

Review Question 4

- When is it more suitable to use a LinkedList over an ArrayList?

Review Question 5

- Which of the above (ArrayList and LinkedList) will be more suitable for:
  - (i) List of goods received; new entries to be added at the end reflecting the date and time of goods receipt.

Review Question 6

- Which of the above (ArrayList and LinkedList) will be more suitable for:
  - (ii) A text editor storing each line as an element in a list; users are allowed to add, delete or alter any part of the document.